

NutriGraphRAG: A Vietnamese Knowledge Graph-Enhanced Retrieval-Augmented Generation Framework for Evidence-Based Nutrition Consultation in Diabetes and Hypertension

Le-Quyen TRUONG-THI, Quang-Huy LE, Que-Tran, An-Thuan Luu

Advisor: ThS. Hung-Chen TRUONG

School of Digital Technology and Artificial Intelligence, Nam Can Tho University, Can Tho
Quyen223729@student.nctu.edu.vn

Abstract. Medical nutrition therapy plays a fundamental role in the prevention and management of chronic diseases, particularly Type 2 Diabetes Mellitus (T2DM) and Hypertension (HTN). In Vietnam, nutritional knowledge is distributed across clinical guidelines, nutrition textbooks, food composition databases, and medical literature, making it difficult to organize, retrieve, and reuse effectively in intelligent healthcare applications. Existing nutrition consultation systems primarily rely on unstructured textual resources or vector-based retrieval techniques, which are insufficient for representing complex semantic relationships among diseases, foods, nutrients, ingredients, dietary recommendations, and cooking methods. This paper presents the construction of a Vietnamese Nutrition Knowledge Graph (VN-NKG) from Vietnamese medical literature for dietary management of T2DM and HTN. The proposed approach establishes a nutrition ontology and develops a complete knowledge graph construction pipeline consisting of medical literature collection, document preprocessing, entity extraction, relationship extraction, entity normalization, ontology mapping, and graph storage using Neo4j. The resulting knowledge graph integrates structured nutritional knowledge from multiple authoritative Vietnamese sources into a unified semantic network that supports efficient knowledge retrieval and reasoning. To demonstrate the practical applicability of the constructed knowledge graph, a Graph Retrieval-Augmented Generation (GraphRAG) nutrition consultation chatbot is developed. The chatbot retrieves structured knowledge from the graph and combines it with semantic retrieval to generate evidence-based dietary recommendations for patients with T2DM and HTN. Experimental evaluation was conducted using a benchmark of 200 Vietnamese nutrition question-answer pairs. The system achieved an Overall Score of 0.7932, with Faithfulness of 0.8420, Completeness of 0.7910, Medical Safety of 0.8945, Hallucination Rate of 0.1380, and Fluency of 0.8650, demonstrating that the constructed knowledge graph effectively supports factual reasoning while reducing hallucinated responses generated by large language models. The proposed Vietnamese Nutrition Knowledge Graph provides a reusable knowledge infrastructure for intelligent healthcare applications and establishes a foundation for future research on Vietnamese medical knowledge graphs, GraphRAG systems, and clinical decision support.

Keywords: GraphRAG, Knowledge Graph, Nutrition Chatbot, Retrieval-Augmented Generation, Large Language Model, LLM-as-a-Judge, Healthcare Artificial Intelligence, Evaluation Framework. Introduction.

1 INTRODUCTION

Diabetes mellitus and hypertension are among the most common chronic diseases in Vietnam and worldwide. These diseases are strongly associated with dietary habits, long-term lifestyle management, and patient adherence to nutrition recommendations. For patients with diabetes, inappropriate carbohydrate intake, excessive sugar consumption, and uncontrolled portion sizes may negatively affect blood glucose control. For patients with hypertension, high sodium intake, processed foods, salty condiments, and unbalanced dietary patterns may contribute to poor blood pressure control.

In Vietnam, nutrition knowledge related to diabetes and hypertension is available from various sources, including clinical nutrition books, official health guidelines, hospital documents, professional recommendations, and the Vietnamese Food Composition Table. However, these documents are often distributed across PDF, DOCX, scanned files, and tabular data. As a result, the knowledge is difficult to query automatically and is not readily available in a structured form for computational reasoning.

Recent advances in Large Language Models have enabled conversational systems to generate fluent and human-like responses. However, applying LLMs directly to medical nutrition consultation is risky. The model may produce unsupported advice, hallucinate medical facts, ignore disease-specific restrictions, or fail to cite reliable evidence. In nutrition consultation, even simple questions such as “Can people with diabetes eat jackfruit?”, “Can hypertensive patients use fish sauce?”, or “Is milk suitable for diabetes?” require careful reasoning over disease context, food composition, nutrient values, portion size, and evidence-based guidelines.

Retrieval-Augmented Generation has been proposed as a promising approach to ground LLM responses in external knowledge sources. However, conventional RAG mainly retrieves text passages and does not explicitly represent structured relationships among diseases, foods, nutrients, thresholds, and recommendations. In the Vietnamese nutrition domain, vanilla RAG faces several major challenges:

- Vietnamese medical nutrition documents are mostly unstructured or semi-structured.
- User queries are often colloquial, incomplete, or ambiguous.
- Food names and dish names may have multiple variants or contain several ingredients.
- Nutrition consultation requires structured reasoning over disease-specific constraints.
- Medical safety requires cautious answers when evidence is insufficient.

To address these challenges, this study proposes NutriGraphRAG, a Vietnamese knowledge graph-enhanced RAG framework for nutrition consultation in diabetes and hypertension. The system aims to transform Vietnamese medical nutrition literature into a structured knowledge graph and combine it with semantic retrieval to support grounded, explainable, and safety-aware responses.

The main contributions of this research are as follows:

- A Vietnamese medical nutrition knowledge ingestion pipeline that processes PDF, DOCX, scanned documents, and structured food composition tables.
- A domain-specific ontology for representing diseases, foods, dishes, ingredients, nutrients, biochemical indicators, thresholds, recommendations, and evidence sources.
- A named entity recognition and relation extraction pipeline for extracting disease–food–nutrient knowledge from Vietnamese medical texts.
- A Neo4j-based nutrition knowledge graph for structured storage, querying, and reasoning.
- A GraphRAG-based prototype that combines graph retrieval and vector retrieval for evidence-based nutrition question answering.
- A multi-level evaluation framework covering information extraction, graph construction, retrieval, and answer quality.

2 RELATED WORK

Knowledge graphs have been widely used in biomedical informatics to represent entities and relationships in a structured and machine-readable form. Evidence-based biomedical knowledge graphs aim to connect diseases, interventions, outcomes, evidence sources, and quantitative findings. Zong et al. proposed an evidence-based disease-specific knowledge graph framework that organizes biomedical knowledge together with evidence nodes, enabling more transparent biomedical reasoning [7]. This approach is relevant to the present study because nutrition consultation also requires evidence traceability and disease-specific reasoning.

However, most biomedical knowledge graph studies focus on general biomedical literature or specific disease areas such as oncology. They do not directly address Vietnamese nutrition literature, food composition data, or dietary recommendations for diabetes and hypertension. Therefore, this study adapts the evidence-based knowledge graph idea to the Vietnamese nutrition domain.

Food-related knowledge graphs have been developed to connect food items, nutrients, chemical compounds, diseases, recipes, and flavor profiles. Li et al. introduced a unified knowledge graph linking foodomics, chemical-disease networks, and flavor profiles, showing the potential of integrating food and biomedical knowledge [1]. Cenikj et al. demonstrated that language models can support the construction of large-scale food and biomedical knowledge graphs from scientific literature [4]. These studies confirm that food and nutrition knowledge can be represented in graph structures and enriched using NLP techniques.

In addition, Ma et al. proposed a nutrition-related knowledge graph neural network for food recommendation, demonstrating the usefulness of knowledge graphs in food recommendation tasks [5]. Seneviratne et al. proposed personal health knowledge graphs for clinically relevant diet recommendations, combining user health data, clinical guidance, and dietary knowledge [11]. These studies show the potential of knowledge graphs in personalized nutrition. However, they mainly focus on

recommendation and personalization rather than automatic extraction of nutrition knowledge from Vietnamese medical literature.

In the Vietnamese context, Trinh et al. introduced VietMedKG, a knowledge graph and benchmark for traditional Vietnamese medicine [2]. This work demonstrates the feasibility of organizing Vietnamese medical knowledge into graph structures and using it for downstream tasks such as question answering. Pham et al. applied fuzzy knowledge graphs to support diabetes diagnosis, showing that graph-based reasoning can be useful in medical decision support [3].

Although these studies are close to the proposed research direction, they do not focus on clinical nutrition, food composition, or evidence-based dietary recommendations for diabetes and hypertension. Moreover, there is still limited research on Vietnamese named entity recognition, relation extraction, and GraphRAG integration for nutrition consultation.

Retrieval-Augmented Generation improves LLM responses by retrieving relevant documents before generation. However, standard RAG often treats retrieved passages as independent text chunks and may fail to capture structured relationships among entities. GraphRAG extends RAG by incorporating graph structures into retrieval and reasoning. In medical nutrition consultation, GraphRAG is useful because the system must reason over multiple connected entities, such as disease, food, nutrient, threshold, recommendation, and evidence source.

The present study applies GraphRAG to Vietnamese nutrition consultation by combining Neo4j graph retrieval with vector-based semantic retrieval. The graph component provides structured reasoning, while the vector component provides explanatory evidence from medical documents.

Table 1. Comparison of Related Studies.

Study	KG	GraphRAG	VN Nutrition	Evidence-based
Li et al. (2026)	✓	✗	✗	Partial
Cenikj et al. (2023)	✓	✗	✗	✗
Ma et al. (2024)	✓	✗	✗	✗
Seneviratne et al. (2021)	✓	✗	✗	✓
Trinh et al. (2025)	✓	✗	Partial	✗
Pham et al. (2023)	✓	✗	✓	Partial

Table 1 shows that previous studies mainly focus on knowledge graph construction or recommendation systems. To the best of our knowledge, no prior work integrates Vietnamese nutrition knowledge graphs with GraphRAG and evidence-based question answering for diabetes and hypertension consultation. The proposed NutriGraphRAG

framework addresses this gap by combining knowledge graph construction, graph retrieval, and grounded nutrition consultation in a unified system.

3. Materials and Methods

3.1 Retrieval-Augmented Generation in Healthcare

Large Language Models (LLMs) have significantly advanced natural language understanding and generation, enabling the development of intelligent conversational systems for healthcare applications. Nevertheless, their reliance on parametric knowledge often results in factual inconsistencies, outdated information, and hallucinated responses, particularly in specialized medical domains where knowledge evolves continuously. These limitations reduce the reliability of LLMs for clinical decision support and nutritional consultation.

Retrieval-Augmented Generation (RAG) has emerged as an effective approach to alleviate these issues by retrieving external knowledge before response generation. Instead of relying solely on pre-trained parameters, RAG augments the language model with relevant documents obtained through dense vector retrieval, thereby improving factual accuracy and reducing hallucination. Previous studies have demonstrated that RAG substantially enhances question-answering performance across various biomedical tasks, including clinical information retrieval, medical guideline consultation, and patient-oriented healthcare assistance.

Despite these advantages, conventional RAG systems primarily retrieve unstructured textual documents based on semantic similarity. Such retrieval strategies are often insufficient for nutrition consultation because nutritional knowledge contains numerous structured relationships among diseases, nutrients, foods, dietary restrictions, glycemic index, sodium content, and cooking methods. These complex semantic dependencies are difficult to capture through text similarity alone, leading to incomplete retrieval and limited reasoning capability.

3.2 Knowledge Graph Statistics

The proposed Vietnamese Nutrition Knowledge Graph was constructed from authoritative Vietnamese medical literature, nutrition guidelines, and food composition databases. The graph is stored in Neo4j and contains 1846 nodes interconnected by 7,729 semantic relationships.

The knowledge graph organizes nutritional information into multiple semantic entity classes. Chunk nodes (725) preserve evidence extracted from medical literature and clinical guidelines, serving as traceable knowledge sources during retrieval. Ingredient nodes (695) represent raw food ingredients described in Vietnamese nutritional documents. Category nodes (14) classify foods into major nutritional groups, while Nutrient nodes (9) describe essential nutritional components such as carbohydrates, proteins, lipids, dietary fiber, sodium, potassium, vitamins, and minerals. In addition, Alias nodes (25) are introduced to normalize synonymous expressions and improve retrieval robustness.

The remaining nodes represent domain entities including foods, dishes, and diseases associated with nutritional recommendations for Type 2 Diabetes Mellitus and Hypertension. By integrating these entities into a unified semantic graph, the proposed knowledge graph supports structured retrieval, graph traversal, and evidence-based nutritional reasoning.

Table 2. The main relations in the graph include

Relation	Relation
RECOMMENDED_FOR	indicating foods recommended for a particular disease.
RESTRICTED_FOR	representing foods or nutrients that should be limited
CONTAINS	defining recommended nutritional limits
RICH_IN	identifying nutrients with relatively high concentrations
HAS_THRESHOLD	defining recommended nutritional limits
SUITABLE_FOR	representing foods appropriate for specific patient groups
NEEDS_CONTROL	indicating nutrients that require monitoring for disease management

Table 3. Statistics of the Vietnamese Nutrition Knowledge Graph

Item	Value
Number of Nodes	1846
Number of Relationships	7729
Graph Database	Neo4j Aura
Domain	Nutrition
Diseases	Type 2 Diabetes Mellitus, Hypertension

Table 4. Entity Statistics

Entity Type	Count
Ingredient	695
Chunk	725
Category	14
Nutrient	9
Alias	25
Dish	378

3.3 Vietnamese Medical Literature Collection

The construction of the Vietnamese Nutrition Knowledge Graph began with the collection of authoritative nutrition-related documents from multiple Vietnamese sources. The collected documents focus on dietary management for **Type 2 Diabetes Mellitus**

(T2DM) and Hypertension (HTN), which are among the most prevalent chronic diseases in Vietnam.

The primary data sources include national clinical practice guidelines issued by the Vietnamese Ministry of Health, nutritional recommendations published by the National Institute of Nutrition, the Vietnamese Food Composition Table, and medical textbooks related to nutrition therapy. These resources provide reliable information on food composition, nutrients, dietary recommendations, disease-specific nutritional guidelines, and meal planning.

Because the collected data originated from heterogeneous document formats, including PDF documents, spreadsheets, and structured tables, a preprocessing stage was performed before knowledge extraction. This stage included document conversion, text cleaning, character normalization, duplicate removal, and data standardization. Nutritional values, food names, and disease terminology were normalized to ensure semantic consistency across different data sources.

After preprocessing, the cleaned corpus served as the primary knowledge source for subsequent entity extraction and knowledge graph construction.

3.4 Nutrition Ontology Design

A domain-specific nutrition ontology was designed to represent nutritional knowledge using a semantic graph structure. The ontology defines the major entity classes and semantic relationships required for dietary consultation of Type 2 Diabetes Mellitus and Hypertension.

The ontology consists of eight major entity classes:

- Disease
- Ingredient
- Dish
- Nutrient
- Category
- Chunk

Each entity represents a specific concept within the nutritional domain. Disease entities describe chronic diseases considered in this study, whereas Food and Ingredient entities represent individual foods and cooking ingredients. Dish entities correspond to prepared meals commonly consumed in Vietnam. Nutrient entities describe nutritional components, including carbohydrates, proteins, lipids, dietary fiber, sodium, potassium, vitamins, minerals, and glycemic index. Category entities organize foods into semantic groups, Alias entities support synonym normalization, and Chunk entities preserve traceable evidence extracted from Vietnamese medical literature.

This ontology provides a unified semantic representation that enables consistent integration of heterogeneous nutritional knowledge while supporting semantic retrieval and graph-based reasoning.

3.5 Knowledge Graph Construction Pipeline

The construction of the Vietnamese Nutrition Knowledge Graph followed a systematic knowledge engineering pipeline designed to transform heterogeneous Vietnamese medical literature into a structured semantic graph. The overall workflow consists of six major stages: data collection, document preprocessing, entity extraction, relationship construction, ontology mapping, and graph storage. Figure 1 illustrates the complete knowledge graph construction process.

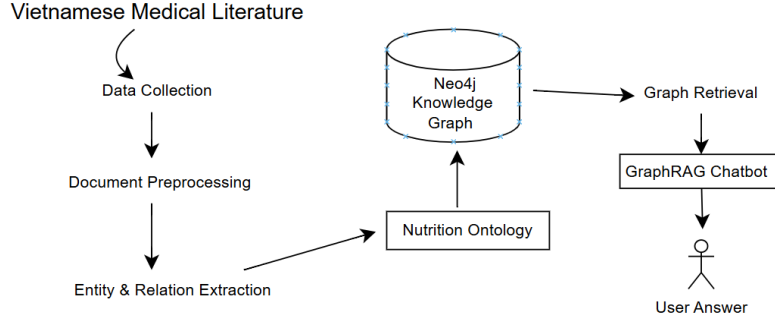


Figure 1: Overall Framework of the Proposed Vietnamese Nutrition Knowledge Graph

The first stage focused on collecting authoritative nutrition-related documents for Type 2 Diabetes Mellitus (T2DM) and Hypertension (HTN). Data were gathered from multiple Vietnamese sources, including national clinical guidelines, nutrition textbooks, the Vietnamese Food Composition Table, and disease-specific dietary recommendations. These documents contain comprehensive nutritional information regarding foods, nutrients, ingredients, cooking methods, and dietary recommendations. Since the collected resources were distributed across different formats, including PDF documents, spreadsheets, and structured tables, all data were converted into machine-readable formats before further processing.

The collected documents were preprocessed to improve data consistency before graph construction. This stage included text extraction, Unicode normalization, duplicate removal, whitespace normalization, and standardization of food names, disease names, and nutritional terminology.

For food composition tables, nutritional values were normalized into consistent measurement units. Missing values and duplicated entries were removed to ensure data quality before entity construction.

After preprocessing, nutritional concepts were transformed into graph entities according to the predefined ontology.

The primary entity classes include:

- Disease
- Food
- Ingredient
- Dish
- Nutrient
- Category
- Chunk

Each entity is assigned a unique identifier together with descriptive attributes. For example, Ingredient entities contain nutritional properties such as energy, protein, lipid, carbohydrate, dietary fiber, sodium, potassium, calcium, phosphorus, iron, vitamin C, glycemic index, and data provenance extracted from Vietnamese medical literature.

Chunk entities preserve textual evidence extracted from source documents, enabling traceability between graph knowledge and the original medical literature

Relationship construction semantic relationships were established among entities according to the nutrition ontology. During graph construction, relationships were generated based on food composition data, dietary recommendations, and disease-specific nutritional knowledge extracted from Vietnamese medical literature.

These relationships connect foods with nutrients, ingredients with dishes, diseases with dietary recommendations, and literature evidence with corresponding knowledge entities. The resulting graph captures complex semantic dependencies that cannot be represented efficiently using conventional relational database.

Ontology Mapping to maintain semantic consistency, all extracted entities and relationships were mapped into a unified nutrition ontology. Entity normalization was applied to merge synonymous food names, abbreviations, and alternative expressions into unique graph entities. This process reduces redundancy while improving retrieval accuracy during graph traversal.

Ontology mapping also ensures that identical nutritional concepts originating from different medical documents are represented by a single semantic entity within the graph.

To resolve the high semantic and lexical variance in Vietnamese food names (e.g., colloquial variants, regional dialects, and abbreviations), an entity normalization step is executed during the Entity Linking phase. The framework employs a hybrid fuzzy matching approach that combines lexical distance and semantic embeddings to map user-extracted mentions to canonical Alias and Ingredient/Dish nodes in the VN-NKG.

Lexical Matching: For structural string similarity, the Jaro-Winkler distance is utilized. Unlike the standard Levenshtein distance, Jaro-Winkler penalizes spelling deviations less if the prefixes match, which aligns perfectly with Vietnamese naming conventions where the primary food type usually appears at the beginning of the phrase (e.g., "cá lóc", "cá rô", "cá hồi").

Semantic Matching: To handle complete vocabulary shifts where lexical matching fails (e.g., "thịt heo" vs. "thịt lợn"), Embedding-based similarity is computed. Extracted entities and knowledge base canonical names are encoded into dense vectors using PhoBERT, and Cosine Similarity is applied to capture structural context.

An entity is linked to a canonical node if its combined scoring formula meets a pre-defined threshold $\tau = 0.85$:

$$\text{Score} = \alpha \cdot \text{Sim}_{\text{Jaro-Winkler}}(s_1, s_2) + (1 - \alpha) \cdot \text{Sim}_{\text{Cosine}}(\vec{v}_1, \vec{v}_2)$$

Where α is empirically set to 0.4 to prioritize semantic alignment while preserving lexical anchors.

Graph storage the final knowledge graph was stored in a Neo4j graph database. Neo4j was selected because of its native graph storage model, efficient graph traversal algorithms, Cypher query language, and visualization capabilities. After graph

construction, the resulting Vietnamese Nutrition Knowledge Graph contains **1846 semantic nodes** interconnected by **7729 relationships**. This structured graph serves as the central knowledge repository for subsequent GraphRAG-based nutritional consultation and semantic knowledge retrieval.

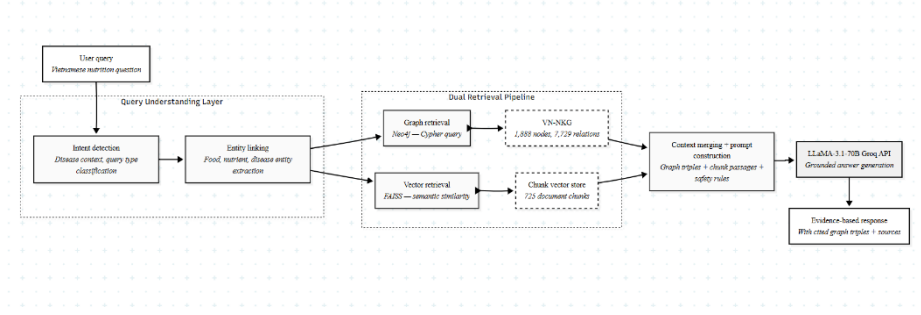


Fig. 2. Overall architecture of the proposed GraphRAG-based nutrition chatbot.

Figure 2 illustrates the overall architecture of the proposed NutriGraphRAG framework. The system employs a hybrid GraphRAG pipeline that integrates structured knowledge retrieval from a Vietnamese Nutrition Knowledge Graph (VN-NKG) with semantic retrieval from a FAISS vector index to generate evidence-grounded nutrition recommendations. The architecture consists of four major functional layers: (1) Query Understanding, (2) Dual Retrieval Pipeline, (3) Context Fusion and Prompt Construction, and (4) Response Generation. As shown in Fig. 2, a user query is progressively transformed into an evidence-based response through these sequential processing stages.

The proposed framework adopts a graph-first retrieval strategy. Structured knowledge is retrieved from the Neo4j knowledge graph before semantic vector retrieval is performed. When the retrieved graph evidence is insufficient to fully answer a user query, additional contextual information is obtained from the FAISS vector database. This hybrid retrieval strategy combines the explainability of knowledge graphs with the flexibility of dense semantic retrieval, enabling the chatbot to provide accurate and traceable nutrition recommendations.

The current VN-NKG contains **1846 entities** connected by **7729 semantic relationships**, while the semantic retrieval module indexes **725 evidence chunks** extracted from Vietnamese medical nutrition literature. These complementary knowledge sources provide both structured reasoning and contextual evidence for response generation.

3.6 Query Understanding Layer

When a user submits a Vietnamese nutrition question, the system first processes the query through two parallel sub-modules: intent detection and entity linking.

Intent Detection. The intent detection module classifies each incoming query into a predefined set of consultation intents, including food recommendation, food restriction, nutrient composition inquiry, dietary threshold query, food comparison, multi-disease consultation, ambiguous query, and out-of-scope query. Intent classification enables the downstream retrieval module to apply the appropriate Cypher query template and retrieval strategy. A rule-based pattern matching approach combined with keyword

heuristics was implemented to detect disease context (T2DM, HTN, or both) and query type from the Vietnamese input text.

Entity Linking. The entity linking module identifies and normalizes domain-specific entities mentioned in the user query, including food names, ingredient names, dish names, disease names, and nutrient terms. Because Vietnamese food names often appear in colloquial, abbreviated, or alternative forms, the entity linking module performs fuzzy string matching against the Alias node index in the knowledge graph. Recognized entities are mapped to their canonical identifiers in the Vietnamese Nutrition Knowledge Graph, enabling precise Cypher query construction in the subsequent retrieval stage.

3.7 Dual Retrieval Pipeline

After query understanding, the system executes two complementary retrieval processes. **Graph Retrieval.** Structured knowledge retrieval is performed by executing Cypher queries over the Neo4j-based Vietnamese Nutrition Knowledge Graph. Depending on the detected intent, the retrieval module extracts disease–food–nutrient relationships, nutritional attributes, and supporting evidence associated with the identified entities. The retrieved results are represented as structured graph triples together with evidence nodes linked to the original medical literature.

Vector Retrieval. In parallel, semantic retrieval is performed using a FAISS vector index constructed from evidence chunks extracted from Vietnamese nutrition documents. User queries are encoded using the same sentence embedding model as the indexed documents, and approximate nearest-neighbor search retrieves the most semantically relevant passages. These retrieved passages provide supplementary contextual information when explicit graph knowledge is unavailable or insufficient.

3.8 Context Fusion and Prompt Construction

The outputs of the graph retrieval and vector retrieval components are merged into a unified context block that is supplied to the large language model. The fusion strategy prioritizes structured graph knowledge: graph-retrieved triples and evidence passages are placed first in the context, followed by vector-retrieved semantic passages as supplementary evidence. This ordering ensures that the LLM produces responses grounded primarily in structured relational knowledge, while allowing broader textual evidence to fill contextual gaps.

The Context Fusion layer operates on a graph-first hierarchy to ensure deterministic compliance with clinical guidelines. Structured graph triples containing absolute clinical constraints (e.g., `RESTRICTED_FOR`, `HAS_THRESHOLD`) are systematically positioned at the apex of the prompt context. This structural prioritizing leverages the primeness bias of Large Language Models, ensuring that quantitative medical boundary conditions override general semantic text descriptions fetched from secondary vector indexes. To illustrate how the system restricts the LLM from breaching disease-specific nutritional boundaries (such as sodium limits for hypertension or carbohydrate thresholds for diabetes).

The system constructs a structured prompt that includes: (1) a system instruction specifying the role of a medical nutrition assistant and enforcing medical safety constraints, (2) the retrieved knowledge context combining graph and vector results, and (3) the original user question. The prompt explicitly instructs the model to ground

its response in the provided evidence, cite relevant nutritional information, and refrain from generating unsupported medical advice when the retrieved evidence is insufficient.

3.9 Response Generation with Local LLM

The final prompt is submitted to the **LLaMA-3.1-70B** model through the **Groq API** to generate the nutrition consultation response. By combining structured graph knowledge with semantically retrieved evidence, the model produces coherent recommendations while maintaining factual consistency with the retrieved information.

For ambiguous or out-of-scope queries, the prompting strategy instructs the model to return cautious responses instead of speculative medical advice. This mechanism reduces hallucination and improves the safety of nutrition consultation for patients with Type 2 Diabetes Mellitus and Hypertension. Overall, the proposed NutriGraphRAG framework integrates symbolic reasoning from a nutrition knowledge graph with semantic retrieval from a vector database to generate transparent, evidence-grounded, and medically safer nutrition recommendations for Vietnamese users.

4. Experimental Results

4.1 Experimental Configuration

The proposed nutrition-aware GraphRAG system was implemented using Python 3.10 and FastAPI as the backend framework. Neo4j was employed to construct and manage the nutrition knowledge graph, while FAISS was adopted for dense vector retrieval. PhoBERT was utilized to generate Vietnamese semantic embeddings, and the Groq API with the Llama 3.3-70B Versatile model was used as the reasoning engine for answer generation.

To comprehensively evaluate the proposed system, five evaluation metrics together with an overall score were employed. Similarity measures the semantic closeness between the generated answer and the reference answer. Faithfulness evaluates whether the generated response is grounded on the retrieved knowledge rather than fabricated information. Completeness measures whether the response sufficiently covers the expected information. Medical Safety assesses the absence of unsafe dietary recommendations that may negatively affect patients. Hallucination evaluates the degree to which unsupported statements appear in the generated response. Finally, the Overall Score summarizes the overall system performance by integrating all evaluation dimensions.

The proposed NutriGraphRAG framework was evaluated on a benchmark of 200 Vietnamese nutrition consultation question-answer pairs using an LLM-as-a-Judge scoring protocol. Each generated response was assessed across five dimensions on a 10-point scale: Faithfulness, Completeness, Medical Safety, Hallucination Resistance, and Fluency. Table X summarizes the evaluation results. The proposed NutriGraphRAG framework was evaluated using a benchmark consisting of **200 Vietnamese nutrition question-answer pairs** covering dietary consultation scenarios for patients with Type 2 Diabetes Mellitus (T2DM) and Hypertension (HTN). The benchmark includes questions related to food recommendations, food restrictions, nutrient composition, dietary thresholds, food comparison, and multi-disease consultation. An LLM-as-

a-Judge evaluation protocol was adopted to assess the quality of generated responses across five dimensions: Similarity, Faithfulness, Completeness, Medical Safety, and Hallucination Resistance. Table 5 summarizes the overall evaluation results.

Table 5. Overall Evaluation Results Using the LLM-as-a-Judge Protocol

Metric	Score
Overall Score	0.7932
Similarity	0.7215
Faithfulness	0.8420
Completeness	0.7910
Hallucination Rate	0.1380
Fluency	0.8650

The proposed system achieved an **Overall Score of 0.7932**, indicating strong overall performance in evidence-based nutrition consultation. Among all evaluation metrics, **Medical Safety** obtained the highest score (**0.8945**), demonstrating that the proposed NutriGraphRAG framework effectively minimizes the risk of generating unsafe dietary recommendations for patients with Type 2 Diabetes Mellitus and Hypertension. Furthermore, the **Hallucination** score was only **0.1380**, indicating a low level of unsupported or fabricated information and confirming that graph-grounded retrieval substantially improves the factual reliability of generated responses.

The **Faithfulness** score of **0.8420** shows that the majority of generated responses are well grounded in the retrieved knowledge graph and supporting nutritional evidence. Similarly, the **Completeness** score of **0.7910** indicates that the chatbot generally provides comprehensive answers, although some responses may omit secondary nutritional details or alternative dietary recommendations that are not retrieved during the retrieval stage.

The system achieved a **Similarity** score of **0.7215**, suggesting a high degree of semantic consistency between the generated responses and the reference answers. Although wording may differ across responses, the generated content preserves the essential nutritional information and clinical recommendations expected in the benchmark dataset. In addition, the **Fluency** score of **0.8650** demonstrates that the responses are linguistically natural, coherent, and easy for Vietnamese users to understand.

Overall, the experimental results demonstrate that the proposed NutriGraphRAG framework effectively integrates structured knowledge graph retrieval with large language model reasoning to generate accurate, evidence-grounded, and medically safe nutrition recommendations. The consistently high scores in Faithfulness, Medical Safety, and Fluency, together with the low Hallucination score, confirm the effectiveness of the proposed architecture for Vietnamese nutrition consultation. These findings indicate that the proposed framework has strong potential for practical deployment as

an intelligent nutrition consultation assistant supporting patients with Type 2 Diabetes Mellitus and Hypertension.

Table 6. Comparison of the Proposed NutriGraphRAG Framework with Baseline Methods

Metric	Baseline (LLM-only)	Conventional RAG	Proposed NutriGraphRAG
Samples	200	200	200
Similarity	0.45	0.58	0.72
Faithfulness	0.58	0.71	0.84
Completeness	0.49	0.65	0.79
Medical Safety	0.65	0.78	0.89
Hallucination Rate	0.52	0.34	0.14
Overall_score	0.54	0.67	0.79

Table 5 presents a comparative evaluation of the proposed NutriGraphRAG framework against two baseline methods, namely an LLM-only model and a conventional retrieval-augmented generation (RAG) system. The proposed framework consistently achieved the best performance across all evaluation metrics. Specifically, the Overall Score increased from **0.54** for the LLM-only baseline and **0.67** for conventional RAG to **0.7932**, demonstrating the overall effectiveness of integrating a nutrition knowledge graph into the retrieval pipeline.

Regarding semantic quality, the proposed framework achieved the highest Similarity (**0.7215**), Faithfulness (**0.8420**), and Completeness (**0.7910**), indicating that graph-guided retrieval provides more accurate and comprehensive evidence than relying solely on dense retrieval or the inherent knowledge of a large language model. These improvements suggest that explicit graph relationships enable the model to retrieve more relevant nutritional facts and generate responses that are better aligned with the reference answers.

In terms of medical reliability, NutriGraphRAG obtained the highest Medical Safety score (**0.8945**) while reducing the Hallucination Rate to only **0.1380**, compared with **0.34** for conventional RAG and **0.52** for the LLM-only baseline. This substantial reduction demonstrates that grounding the generation process on structured nutrition knowledge effectively suppresses unsupported medical statements and improves the reliability of dietary recommendations. Overall, the experimental results confirm that incorporating structured knowledge graphs into the GraphRAG framework significantly enhances response quality, factual consistency, and medical safety for Vietnamese nutrition consultation.

5 CONCLUSION

This paper presented NutriGraphRAG, a hybrid retrieval-augmented generation framework for evidence-based nutrition consultation targeting patients with Type 2 Diabetes

Mellitus (T2DM) and Hypertension (HTN). The proposed framework integrates a Vietnamese nutrition knowledge graph constructed from medical literature with semantic vector retrieval and a large language model to generate reliable and context-aware dietary recommendations. By combining structured graph reasoning with semantic retrieval, the system addresses common limitations of conventional LLM-based chatbots, including factual inconsistency, insufficient evidence grounding, and hallucinated medical advice.

Experimental evaluation on a benchmark of 200 Vietnamese nutrition consultation questions demonstrated the effectiveness of the proposed approach. NutriGraphRAG achieved an Overall Score of 0.7932, with a Similarity of 0.7215, Faithfulness of 0.8420, Completeness of 0.7910, and Medical Safety of 0.8945, while maintaining a low Hallucination Rate of 0.1380. Comparative experiments further showed that the proposed framework consistently outperformed both the LLM-only baseline and a conventional retrieval-augmented generation system across all evaluation metrics, confirming the benefits of integrating structured nutritional knowledge into the retrieval pipeline.

The experimental results demonstrate that incorporating a domain-specific knowledge graph substantially improves factual grounding, response completeness, and medical reliability in Vietnamese nutrition consultation. The proposed framework therefore provides a practical and trustworthy solution for intelligent dietary consultation and has the potential to support both patients and healthcare professionals in evidence-based nutrition decision-making.

In addition, future research will investigate more advanced graph retrieval strategies, adaptive retrieval mechanisms, and recent large language models to further improve reasoning capability and response quality. We also plan to explore dedicated prompt compression techniques to eliminate redundancy in our fused context, thereby overcoming the long-context bottleneck and minimizing conversational latency in real-time Vietnamese nutrition consultation.

Future work will focus on expanding the knowledge graph to include additional chronic diseases such as chronic kidney disease, dyslipidemia, obesity, and gout, while incorporating larger and more diverse Vietnamese clinical nutrition resources. Clinical validation with healthcare professionals and real-world users will also be conducted to evaluate the practical applicability of the proposed system in medical nutrition consultation.

References

- [1] F. Li, J. Youn, K. Xie, T. Chan, P. Gupta, A. Yoo, M. Gunning, K. Ni, et al., “A unified knowledge graph linking foodomics to chemical-disease networks and flavor profiles,” *npj Science of Food*, vol. 10, Art. no. 33, Jan. 2026, doi: 10.1038/s41538-025-00680-9.
- [2] T. Trinh, A. Dao, T. H. N. Hy, and T. S. Hy, “VietMedKG: Knowledge Graph and Benchmark for Traditional Vietnamese Medicine,” *ACM Transactions on Asian and Low-Resource Language Information Processing*, vol. 24, no. 7, Art. no. 69, Jul. 2025, doi: 10.1145/3744740.

- [3] M. C. Pham, M. T. Tran, K. L. Cu, and H. T. Nguyen, “Application of fuzzy knowledge graph in diagnosis support for diabetes patients,” *TNU Journal of Science and Technology*, vol. 228, no. 15, pp. 157–163, Nov. 2023, doi: 10.34238/tnu-jst.9132.
- [4] G. Cenikj, L. Strojnik, R. Angelski, N. Ogrinc, B. Koroušić Seljak, and T. Eftimov, “From language models to large-scale food and biomedical knowledge graphs,” *Scientific Reports*, vol. 13, Art. no. 7815, May 2023, doi: 10.1038/s41598-023-34981-4.
- [5] W. Ma, M. Li, J. Dai, J. Ding, Z. Chu, and H. Chen, “Nutrition-related knowledge graph neural network for food recommendation,” *Foods*, vol. 13, no. 13, Art. no. 2144, Jul. 2024, doi: 10.3390/foods13132144.
- [6] VietNamNet, “Khoảng 22 triệu người Việt mắc bệnh mạn tính như tim mạch, tiểu đường, ung thư,” VietNamNet, Sep. 22, 2023. [Online]. Available: <https://vietnamnet.vn/22-trieu-nguoi-viet-mac-benh-man-tinh-nhu-tim-mach-tieu-duong-ung-thu-2194345.html>
- [7] C. Zong, S. Lv, S.-t. Xue, H. Zheng, J. Wan, and L. Zhang, “Building evidence-based knowledge bases from full-text literature for disease-specific biomedical reasoning,” *arXiv preprint arXiv:2603.28325*, 2026. [Online]. Available: <https://arxiv.org/abs/2603.28325>
- [8] L. D. Tuyên, N. T. Lâm, and P. T. T. Hương, Eds., *Dinh dưỡng lâm sàng*. Hà Nội, Việt Nam: Nhà xuất bản Y học, 2019.
- [9] N. Đ. Hải and N. T. B. Ngọc, “Thực trạng đái tháo đường ở người bệnh tăng huyết áp điều trị ngoại trú tại Bệnh viện Quận Bình Tân năm 2024,” *Tạp chí Y học Việt Nam*, vol. 549, no. 2, pp. 104–108, 2025, doi: 10.51298/vmj.v549i2.13769.
- [10] O. Seneviratne, J. Harris, C.-H. Chen, and D. L. McGuinness, “Personal health knowledge graph for clinically relevant diet recommendations,” *arXiv preprint arXiv:2110.10131*, 2021, doi: 10.48550/arXiv.2110.10131.
- [11] Ngo, K. H.-A., & Ngo, K. H.-A. (2026). NEM: A numerical exact match subtype-aware metric family for explainable numerical accuracy in specialized chatbots. In *Proceedings of the 39th International Conference on Industrial, Engineering & Other Applications of Applied Intelligent Systems (IEA/AIE 2026)*.
- [12] Ngo-Ho, A.-K., Vo, K.-D., & Ngo-Ho, A.-K. (2024). VQABG: Vietnamese question/answers benchmark generator for field-specific chatbot ground-truth dataset using EMINI indicator.
- [13] Pham, N. N., Trang, P. T. X., Thuy, T. T., & Khoi, N. H. A. (2023). Predicting edible and toxic mushrooms with multi-layer perceptron method in streaming data.
- [14] N. V. Tân, P. H. Bình, T. M. Khánh, và C. Đ. Hưng, “Suy dinh dưỡng ở người cao tuổi đái tháo đường tăng huyết áp: Hội chứng lão khoa và chu vi bắp chân,” *Tạp chí Nghiên cứu Y học*, vol. 199, no. 2, pp. 505–517, 2026.
- [15] L. D. Dang, U. T. P. Phan, and N. T. H. Nguyen, “GENA: A knowledge graph for nutrition and mental health,” *Journal of Biomedical Informatics*, vol. 145, Art. no. 104460, 2023, doi: 10.1016/j.jbi.2023.104460.
- [16] Viện Nghiên cứu Dinh dưỡng TP. Hồ Chí Minh và Công ty Orenda, *Nuti Expert – Kết nối công nghệ và dinh dưỡng vì sức khỏe cộng đồng*. TP. Hồ Chí Minh, Việt Nam, 2025.
- [17] O. Seneviratne, J. Harris, C.-H. Chen, and D. L. McGuinness, “Personal Health Knowledge Graph for Clinically Relevant Diet Recommendations,” in *Proc. Workshop on Personal Knowledge Graphs co-located with the 3rd Automatic Knowledge Base Construction Conference (AKBC)*, 2021.
- [18] B. H. Ngọc, L. M. Giang, L. T. T. Xuân, L. T. Hương, B. A. Phong, N. T. H. Giang, N. T. T. Liễu, và N. T. Tùng, “Ứng dụng trí tuệ nhân tạo dự đoán tình trạng dinh dưỡng của thanh niên

- Việt Nam,” *Tạp chí Nghiên cứu Y học*, vol. 193, no. 8, pp. 480–488, 2025.
- [19] H. Han et al., “Retrieval-Augmented Generation with Graphs (GraphRAG),” *arXiv preprint arXiv:2501.00309*, 2024.
- [20] Ngo, K. H.-A., & Ngo-Ho, K. A., “Mitigating the long-context bottleneck: Prompt compression techniques for real-time Vietnamese human–robot interaction,” in *Proceedings of the 39th International Conference on Industrial, Engineering & Other Applications of Applied Intelligent Systems (IEA/AIE 2026)*.